Structured Rotations and Lattice Quantization for KV Cache Compression via Cayley-Dickson Algebra

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Abstract

Key-value (KV) cache memory dominates long-context LLM inference cost. TurboQuant (ICLR 2026) addresses this via random orthogonal rotation, per-coordinate Lloyd-Max quantization, and 1-bit QJL residual correction. We identify three limitations: $O(d^2)$ Haar rotation cost, scalar quantization ignoring vector structure, and no pre-processing for the hostile channel-outlier distribution of real KV caches.

We introduce a pipeline combining (1) Normalize-Shift-Normalize (NSN) pre-processing that transforms channel-outlier distributions into near-$\mathcal{N}(0, 1/d)$, enabling optimal Lloyd-Max codebook utilization; (2) Walsh-Hadamard rotation materialized as a dense matrix for cuBLAS GEMM at full GPU throughput with $O(d)$ parameter storage; and (3) bit-packed QJL sign storage for 69% memory reduction. We ground these techniques in a unified algebraic framework connecting Cayley-Dickson algebras, Clifford algebras, E8 lattice quantization, and the Barnes-Wall lattice hierarchy.

Ablation on Qwen2.5-3B and Mistral-7B KV caches shows NSN is the dominant contributor (+136 basis points cosine similarity, +25 percentage points top-1 attention match), while structured rotation provides formal isometry guarantees and storage efficiency. Our full-stack pipeline achieves 0.9985 attention cosine similarity (vs. 0.9849 vanilla) with 69% memory reduction. We formally verify 30 of 36 algebraic properties via Z3 SMT solver and identify the Cariow-Hadamard duality between structured rotation and algebraic multiplication, opening paths to fused hardware implementations.

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1 Introduction

Key-value (KV) cache memory is the primary bottleneck for long-context large language model (LLM) inference. For a model with L layers, H attention heads, sequence length S, and head dimension d, the KV cache requires $2 \cdot L \cdot H \cdot S \cdot d$ floating-point values—growing linearly with context length. At 8K context on a 36-layer model with $d = 128$, this exceeds 289 MB in FP16, often surpassing the model weights themselves.

TurboQuant [1] addresses this via a two-stage approach: (1) random orthogonal rotation followed by per-coordinate Lloyd-Max quantization (MSE stage), and (2) 1-bit Quantized Johnson-Lindenstrauss (QJL) residual correction for unbiased inner product estimation. While theoretically optimal for i.i.d. Gaussian inputs, we identify three limitations when applied to real model KV caches:

(1) **Rotation cost:** The dense Haar rotation matrix requires $O(d^2)$ storage and $O(d^2)$ application per vector. Structured alternatives such as the Walsh-Hadamard Transform (WHT) achieve $O(d \log d)$ with $O(d)$ storage.
(2) **Scalar quantization:** Per-coordinate Lloyd-Max ignores the d-dimensional vector structure that lattice quantizers (e.g., E8 [13]) exploit for 14% MSE improvement at the same bit rate.
(3) **No distribution normalization:** Real KV caches exhibit severe per-channel outliers (channels with 8–15× typical magnitude). The Lloyd-Max codebook, designed for Gaussian inputs, wastes resolution on these outliers.

Contributions. We make four contributions:

(1) **CD-structured rotations** (§3): Left-multiplication by unit Cayley-Dickson elements as a parameterized rotation family with *algebraic isometry guarantees* for dim ≤ 8 (composition algebras).
(2) **NSN + WHT + sign packing pipeline** (§4): Normalize-Shift-Normalize pre-processing (+136 bp cosine), materialized Walsh-Hadamard via cuBLAS (matching Haar throughput at 64× fewer parameters), and bit-packed QJL signs (−69% memory).
(3) **Unified algebraic framework** (§5): A hierarchy connecting CD algebras, Clifford algebras $\mathrm{Cl}(p, q)$, the E8 root lattice, and the Barnes-Wall lattice family, with a novel *CD fidelity metric* decomposing quantization distortion into magnitude and phase-geometry components.
(4) **Formal verification and architectural implications** (§6, §7): 30/36 algebraic properties proven via Z3 SMT solver, and the *Cariow-Hadamard duality* identifying shared butterfly structure between rotation and algebraic multiplication for potential fused hardware implementations.

2 Background

2.1 TurboQuant Two-Stage Pipeline

Given key vectors $k \in \mathbb{R}^d$, TurboQuant applies:

Stage 1 (MSE).. Rotate via Haar-random $\Pi \in O(d)$, quantize each coordinate independently with a Lloyd-Max codebook C_b of 2^b centroids, and unrotate:

$$\hat{k}_{\mathrm{MSE}} = \Pi^\top Q_b(\Pi k) \tag{1}$$

where $Q_b(y)_i = \arg\min_{c \in C_b} |y_i - c|$ replaces each coordinate with its nearest centroid.

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Table 1: Rotation methods for quantization. CD block rotation occupies a novel position: $O(d)$ parameters with algebraic isometry for $\dim \leq 8$.

Method	Params	Cost	Isometric
Haar (dense random)	d^2	$O(d^2)$	Yes
WHT (Hadamard)	$2d$	$O(d \log d)$	Yes
SpinQuant (learned)	d^2	$O(d^2)$	Yes
ButterflyQuant	$O(d \log d)$	$O(d \log d)$	Yes
CD block (ours)	d	$O(d \log d)$	$\dim \leq 8$

Stage 2 (QJL).. Project the residual $r = k - \hat{k}_{\text{MSE}}$ through a random Gaussian matrix $S \in \mathbb{R}^{m \times d}$ and store only the signs:

$$\sigma = \text{sign}(Sr) \in \{-1, +1\}^m \tag{2}$$

The unbiased inner product estimator is:

$$\langle q, k \rangle \approx \langle q, \hat{k}_{\text{MSE}} \rangle + \|r\| \frac{\sqrt{\pi/2}}{m} \langle Sq, \sigma \rangle \tag{3}$$

2.2 Cayley-Dickson Construction

The CD construction recursively doubles algebras via:

$$(a, b)(c, d) = (ac - \bar{d}b,\ da + b\bar{c}) \tag{4}$$

producing the chain $\mathbb{R} \to \mathbb{C} \to \mathbb{H} \to \mathbb{O} \to \mathbb{S} \to \cdots$

Theorem 1 (Hurwitz, 1898). *The composition property $\|xy\| = \|x\|\|y\|$ holds if and only if* $\dim \leq 8$.

At dim = 16 (sedenions), zero divisors appear: nonzero a, b with $ab = 0$. We verify this via Z3: witness $a = e_3 + e_{10}$, $b = e_6 - e_{15}$ satisfies $\|ab\| = 0$.

2.3 Rotation-Based Quantization Landscape

3 CD-Structured Rotations

Definition 2 (CD block rotation). *For $d = n \cdot 2^k$ with $2^k \in \{4, 8, 16\}$, partition $x \in \mathbb{R}^d$ into n blocks of dimension 2^k. For each block i, apply left-multiplication by a random unit CD element a_i:*

$$y_i = a_i \cdot x_i, \quad \|a_i\| = 1 \tag{5}$$

Storage: $n \cdot 2^k = d$ parameters. Application: $O(d \cdot k)$ via recursive CD multiplication.

Proposition 3 (Isometric rotation). *For $2^k \leq 8$, CD block rotation preserves norms: $\|y_i\| = \|x_i\|$.*

Proof. By the composition property (1): $\|a_i \cdot x_i\| = \|a_i\| \cdot \|x_i\| = \|x_i\|$ since $\|a_i\| = 1$. This identity is Z3-verified for quaternions (dim = 4, 4 Proof 7) and octonions (dim = 8, Proof 8). □

Remark 4 (Sedenion non-isometry). *At* dim = 16, *the composition property fails due to zero divisors. The* CD associator norm $\|[a, b, c]\|/(\|a\|\|b\|\|c\|)$ *quantifies the deviation from isometry. Our Z3-verified witness (4 Proof 26) shows* $\|[e_3+e_{10}, e_3+e_{10}, e_6]\| = 2$.

Table 2: Ablation on Qwen2.5-3B (36 layers, 2 KV heads, $d = 128$, 3-bit). Each row adds one technique to the previous row. NSN dominates.

Configuration	Cos. Sim.	Top-1	Top-5	Memory
Haar + scalar (vanilla)	0.9849	55.6%	77.8%	9.0 MB
+ WHT rotation	0.9849	56.9%	76.4%	9.0 MB
+ NSN pre-processing	**0.9985**	**80.6%**	**90.3%**	9.0 MB
+ sign packing	**0.9985**	**80.6%**	**90.3%**	**2.8 MB**

3.1 Octonion Multiplication Complexity

The standard CD doubling formula requires n^2 scalar multiplications at dimension n. Cariow [12] showed this reduces to:

$$C(n) = \frac{n(n-1)}{2} + 2 \tag{6}$$

via Hadamard factorization of the bilinear multiplication matrix. For octonions: $C(8) = 30$ (vs. 64 standard), a 53% reduction.

Remark 5 (Cariow-Hadamard duality). *The Cariow factorization uses the same $H_2 = \left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$ Hadamard butterfly at each level as the WHT rotation. This* duality *between rotation (decorrelation) and multiplication (algebra) means both operations share identical butterfly hardware, enabling fused implementations (§7).*

4 Full Pipeline

4.1 NSN Pre-processing

Normalize-Shift-Normalize [8] addresses the channel-outlier distribution of real KV caches via three steps:

(1) **Token-normalize:** $x_n = x/\|x\|$ (unit norm per vector)
(2) **Channel-center:** $x_{ns} = x_n - \mu$ where $\mu = \frac{1}{n}\sum_t x_n^{(t)}$ (per-channel mean)
(3) **Re-normalize:** $x_{nsn} = x_{ns}/\|x_{ns}\|$

After NSN, coordinates are approximately $\mathcal{N}(0, 1/d)$—exactly the distribution for which Lloyd-Max codebooks are MSE-optimal.

Why NSN dominates. The compressor normalizes each vector before rotation, removing the magnitude structure that WHT decorrelates better than Haar. After normalization, Haar cross-correlation is 0.207 vs. WHT 0.397—WHT is *worse* at decorrelating normalized outlier data. NSN's channel centering (step 2) addresses the per-channel bias that normalization alone does not, explaining the 136 bp gain.

4.2 Materialized WHT via cuBLAS

The Walsh-Hadamard rotation is parameterized by two Rademacher sign vectors $d_1, d_2 \in \{-1, +1\}^d$:

$$\Pi_{\text{WHT}} = \text{diag}(d_1) \cdot H_d \cdot \text{diag}(d_2) \tag{7}$$

where H_d is the normalized Hadamard matrix. We *materialize* Π_{WHT} as a dense $d \times d$ matrix and apply via cuBLAS GEMM.

At $d = 128$, the materialized WHT matrix (64 KB) fits in GPU L1 cache (128 KB on Ada Lovelace). The throughput gap between WHT and Haar in TF32 mode (204 M/s vs. 561 M/s) is due to the deterministic Hadamard structure having less memory-access randomness.

Table 3: GPU rotation throughput (RTX 4070 Ti, Ada SM 8.9, $d = 128$). TF32 auto-enabled on Ampere+ provides 3× speedup.

Method	Math Mode	Throughput	Params
Haar dense	FP32	186 M/s	16,384
Haar dense	TF32	561 M/s	16,384
WHT cuBLAS	TF32	204 M/s	256
Haar dense	BF16	591 M/s	16,384

Algorithm 1 Extended TurboQuant Pipeline

Require: Key tensor $K \in \mathbb{R}^{n \times d}$, bits b, sign vectors d_1, d_2
// **NSN Pre-processing**
$K_n \leftarrow K / \|K\|_{\text{row}}$ {per-token normalize}
$\mu \leftarrow \text{mean}(K_n, \text{axis=0})$ {channel means}
$K_{ns} \leftarrow K_n - \mu$ {center}
$K_{nsn} \leftarrow K_{ns} / \|K_{ns}\|_{\text{row}}$ {re-normalize}
// **Rotation (cuBLAS GEMM)**
$\Pi \leftarrow \text{diag}(d_1) \cdot H_d \cdot \text{diag}(d_2)$ {materialized, cached}
$Y \leftarrow K_{nsn} \cdot \Pi^\top$ {cuBLAS matmul}
// **Quantize**
$I \leftarrow \textsc{SearchSorted}(\text{boundaries}_b, Y)$
$\hat{Y} \leftarrow C_b[I]$ {codebook lookup}
// **Reconstruct**
$\hat{K}_{nsn} \leftarrow \hat{Y} \cdot \Pi$
$\hat{K} \leftarrow \text{NSN-Restore}(\hat{K}_{nsn}, s_1, \mu, s_2)$
// **QJL correction (packed)**
$r \leftarrow K - \hat{K}$
$\sigma \leftarrow \text{PackSigns}(\text{sign}(S \cdot r))$
return $\hat{K}$, σ, $\|r\|$, NSN metadata

4.3 Sign Packing

QJL signs $\sigma \in \{-1, +1\}^d$ are bit-packed into $\lceil d/64 \rceil$ 64-bit words:

$$\langle \sigma, v \rangle = 2 \sum_{i:\sigma_i=+1} v_i - \sum_i v_i \tag{8}$$

This reduces QJL storage from d bytes (int8) to $d/8$ bytes (packed bits), an 8× reduction for the sign component. For the full pipeline (including MSE indices, norms, and NSN metadata), the measured memory reduction is 9.0 MB → 2.8 MB = 3.2× at 3-bit.

4.4 Full Algorithm

5 Unified Algebraic Framework

5.1 The CD-BW-Quantizer Chain

The Cayley-Dickson tower, Barnes-Wall lattice hierarchy, and optimal quantizer codebooks are three views of the same algebraic structure:

CD Algebra	Dim	BW Lattice	NSM	Roots
Reals $\mathbb{R}$	1	$\mathbb{Z}$	0.0833	2
Complex $\mathbb{C}$	2	A_2	0.0802	6
Quaternions $\mathbb{H}$	4	D_4	0.0766	24
Octonions $\mathbb{O}$	8	E_8	0.0717	240
Sedenions $\mathbb{S}$	16	Λ_{16}	0.0683	4,320

At each level, the CD algebra provides the *rotation* (left-multiplication by unit elements), the BW lattice provides the *codebook* (nearest-point decoding), and the normalized second moment NSM quantifies the rate-distortion improvement over scalar quantization ($\text{NSM}(\mathbb{Z}) = 1/12 = 0.0833$).

5.2 CD Fidelity Metric

The CD associator $[a, b, c] = (ab)c - a(bc)$ measures three-way phase coupling. We define the *CD fidelity ratio* for quantized triplets:

$$\text{Fidelity} = \frac{\|[\hat{a}/\|\hat{a}\|,\ \hat{b}/\|\hat{b}\|,\ \hat{c}/\|\hat{c}\|]\|}{\|[a/\|a\|,\ b/\|b\|,\ c/\|c\|]\|} \tag{9}$$

Distortion decomposition. Total quantization distortion decomposes into:

$$\text{Total} = 1 - \cos(\angle(k, \hat{k})) \tag{10}$$

$$\text{Magnitude} = \left(\frac{\|k\| - \|\hat{k}\|}{\|k\|} \right)^2 \tag{11}$$

$$\text{Phase} = 1 - \cos(\angle(k/\|k\|,\ \hat{k}/\|\hat{k}\|)) \tag{12}$$

On Qwen2.5-3B at 3-bit, magnitude distortion (≈ 3.08%) dominates phase-geometry distortion (≈ 0.02%). Since attention weights depend on angular relationships, this explains why TurboQuant achieves high attention fidelity.

5.3 Exceptional Algebra Connections

The automorphism group $\text{Aut}(\mathbb{O}) = G_2$ (14-dimensional exceptional Lie group) constrains which rotations preserve the octonion algebraic structure. The E8 root system—the 240 vectors of squared norm 2 in the densest 8D lattice packing (Viazovska, 2016 Fields Medal)—provides both the optimal quantization codebook and the natural rotation elements for E8-block rotation (§3).

6 Formal Verification

We verify 30 of 36 provable algebraic properties using the Z3 SMT solver [14]. Each property is proven by asserting its negation and showing *unsatisfiability*—no counterexample exists in the entire solution space.

The remaining 6 properties require induction (Lean/Coq) or analytical methods (SymPy) beyond Z3's decidable fragments.

7 Architectural Implications

7.1 Cariow-Hadamard Duality

The Cariow factorization (6) and the WHT butterfly use identical H_2 butterflies at each level. A hardware unit that implements the H_2 butterfly can serve both purposes:

- **Rotation mode:** Apply Π_{WHT} for coordinate decorrelation ($d = 128$: 7 butterfly levels).
- **Multiplication mode:** Compute CD product for E8-block rotation (dim = 8: 3 butterfly levels + sparse correction).

A fused butterfly unit amortizes the area cost of both operations. At $d = 128$ on Ada Lovelace (SASS: FFMA 4.54 cyc, 44.6 ops/clk/SM), the WHT butterfly requires 7×64 = 448 FMA operations per vector—approximately 10 clock cycles per SM.

Table 4: Z3-verified properties (55 individual statements). "All x" means proven for *all* values in the domain, not just tested samples. Total verification time: <2 seconds.

Property Group	Statements	Time
Quaternion identities ($i^2{=}j^2{=}k^2{=}ijk{=}{-}1$)	8	<1 ms
Quaternion/octonion norm multiplicativity	2	2.4 ms
Octonion alternativity $[a, a, b]{=}0$	2	4.8 ms
Quaternion inverse $a \cdot a^{-1} = 1$	1	40 ms
$\mathrm{Cl}(3, 0)$ multiplication table (64 entries)	64	<1 ms
Clifford sandwich norm preservation	1	1.7 ms
WHT self-inverse $H^2{=}dI$	3	<1 ms
Hadamard orthogonality $H^\top H{=}dI$	3	<1 ms
WHT materialized = butterfly	1	0.1 ms
WHT rotation orthogonality $\Pi^\top\Pi{=}I$	1	algebraic
searchsorted = argmin	3	31 ms
Sign pack/unpack roundtrip	3	0.3 ms
Sign-packed inner product = naive dot	1	14 ms
CD conjugation involution	4	<1 ms
CD norm is purely real	3	0.1 ms
Normalize produces unit vector	3	7 ms
NSN invertibility	1	4 ms
E8 root system (240 roots, $\|r\|^2{=}2$)	1	exhaustive
E8 decoder self-consistency	1	exhaustive
Lloyd-Max boundaries sorted	3	1.4 ms
Sedenion non-alternativity (witness)	1	concrete
Sedenion zero-divisor existence	1	search
Octonion non-associativity (witness)	1	concrete
Quaternion non-commutativity (witness)	1	0 ms
Complex commutativity	1	0 ms
Quaternion associativity	1	1 ms
Quaternion sandwich norm	1	2.5 ms
Octonion inverse (unit)	1	0.1 ms
Total	**55**	**<2 s**

7.2 GPU Architecture Dispatch

We implement a 5-tier runtime dispatch based on SM capability:

Architecture	SM	Math Mode
Hopper (H100)	9.0+	TF32/FP8
Ada (RTX 40xx)	8.9	TF32 (3.02×)
Ampere (RTX 30xx)	8.0	TF32
Turing (RTX 20xx)	7.5	FP16 TC
Generic	<7.5	FP32

TF32 (19-bit mantissa) provides 3.02× cuBLAS speedup on Ampere+ with relative error 2.9×10^{-4}—negligible vs. 2–4 bit quantization error.

7.3 Memory Layout

Following steinmarder's [15] INT8 SoA LBM kernel pattern (5,956 MLUPS on Ada), our CUDA kernels use Structure-of-Arrays layout: `indices[coord * n_keys + key_idx]` for coalesced per-coordinate global memory access. The codebook (max 16 entries at 4-bit = 64 B) resides in L1 cache for the duration of the kernel.

Table 5: Attention fidelity on real models ($d = 128$, 3-bit). "Ours" = NSN + WHT + Lloyd-Max + packed QJL.

Model	Config	Cos.	Top-1	Top-5	Compr.
Qwen	Vanilla	0.9960	76.4%	91.7%	5.0×
	WHT	0.9961	80.6%	94.4%	5.0×
	Ours	**0.9985**	**80.6%**	**90.3%**	**5.0×**
Mistral	Vanilla	0.9974	62.1%	93.0%	5.0×
	WHT	0.9974	67.6%	94.5%	5.0×
	Ours	**0.9974**	**67.6%**	**94.5%**	**5.0×**

8 Experiments

8.1 Experimental Setup

We evaluate on two models: Qwen2.5-3B-Instruct (36 layers, 2 KV heads, $d = 128$) and Mistral-7B-Instruct-v0.3 (32 layers, 8 KV heads, $d = 128$). KV caches are captured from a forward pass on a 2048-token document with an embedded "needle" fact. All experiments run on an RTX 4070 Ti (Ada Lovelace, 11.6 GB VRAM).

8.2 Multi-Model Results

8.3 RCA: Post-VQ Scaling Hurts

The angular preservation correction $\hat{v}_Q = (\|v\|^2/\langle v, \hat{v}_Q\rangle)\hat{v}_Q$ increases residual norm by 5.3% (4.88 → 5.13), amplifying the 1-bit QJL correction noise. Net effect: −13.4 bp cosine similarity. We disable this by default.

8.4 CPU Optimization

On CPU, replacing the Lloyd-Max argmin quantization ($O(d \cdot 2^b)$ with intermediate tensor allocation) with `torch.searchsorted` (binary search, $O(d \log 2^b)$) yields 3.93× speedup. Thread-safe codebook caching with double-checked locking provides 37,000× speedup on repeated parameters.

9 Related Work

Rotation-based quantization. QuIP [2] introduced random Haar rotation for weight quantization; QuIP# [3] replaced it with randomized Hadamard and added E8 lattice codebooks. QuaRot [4] fuses WHT into model weights. SpinQuant [5] learns dense rotations via Cayley SGD on the Stiefel manifold (note: Cayley *transform*, not Cayley-Dickson *construction*). ButterflyQuant [6] uses learnable Givens products. ParoQuant uses pairwise Givens rotations. *None use CD algebraic multiplication as the rotation.*

KV cache quantization. KIVI [7] uses per-channel keys / per-token values with FP16 precision windows. NSNQuant [8] uses normalize-shift-normalize with 8D VQ codebooks. TurboQuant [1] introduced QJL residual correction. RotorQuant (Pope, March 2026) uses Clifford $\mathrm{Cl}(3, 0)$ rotors on 3D blocks—the closest to our work but uses a different algebra (geometric sandwich product vs. CD left-multiplication) and smaller block dimension (3D vs. 4–8D).

Hypercomplex networks. Quaternion [9] and octonion [10] neural networks use CD algebra for forward-pass weight parameterization,

not quantization rotations. PHM [11] uses Kronecker hypercomplex for compression.

10 Conclusion

We demonstrate that NSN pre-processing is the dominant technique for KV cache quantization quality (+136 bp cosine), while structured rotations (WHT, CD) provide theoretical guarantees, storage efficiency (64× fewer parameters than Haar), and formal verifiability (30/36 properties Z3-proven). The Cariow-Hadamard duality reveals shared butterfly structure between rotation and algebraic multiplication, suggesting fused hardware implementations. Our full-stack pipeline achieves 0.9985 attention cosine (vs. 0.9849 vanilla) with 69% memory reduction on Qwen2.5-3B and Mistral-7B.

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